

Internet of Things

Energy Balance Protocols

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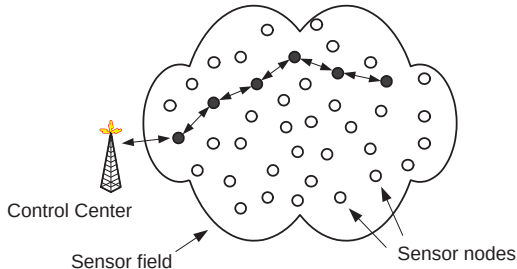
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A “canonical” problem: Local Event Detection and Data Propagation

A single sensor, p , senses a **local event** $\mathcal{E}$. The general **propagation problem** is the following:

“How can sensor p , via cooperation with the rest of the sensors in the network, propagate information about event $\mathcal{E}$ to the control center?”





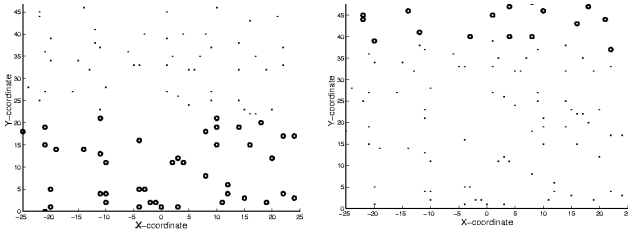
Representative data propagation protocols

- **Directed Diffusion (DD):** a *tree-structure* protocol (suitable for low dynamics)
- **LEACH:** *clustering* and *probabilistic approach* (suitable for small area networks)
- **Local Target Protocol (LTP):** *greedy, single-path optimization* (best for dense networks)
- **Probabilistic Forwarding Protocol (PFR):** *redundant optimized* transmissions (good efficiency / fault-tolerance trade-offs, best for sparse networks)
- **Energy Balance Protocol (EBP):** guaranteeing *same per sensor energy* (prolongs network life-time)



The Energy Balance Protocol (EBP)

All protocols *tend to “strain” some specific nodes* in the network.



- In a *hop-by-hop* scheme the nodes *closer to the sink* tend to be overused.
- In a *direct transmission* scheme the *distant nodes* tend to be overused.



Research Question

“How can we achieve *equal energy dissipation per node* in order to prolong the network lifetime by avoiding early network disconnection?”



A Probabilistic Solution

- Direct transmission cost is much larger than that of one-hop transmission and exhausts distant nodes.
- One-hop transmissions are cheap but tend to overuse nodes that lie closer to the sink.

Solution: “Each node *chooses randomly* whether to propagate the event *one-hop closer to the sink* or to transmit it *directly to the sink*”.

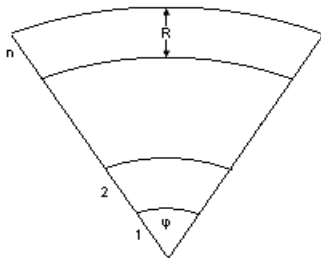
The *goal* is to “*balance*” the two types of transmissions and achieve *equal average energy dissipation per sensor*.



The EBP Protocol (I)

- **Network Partition:**

- *Partition* the network into n sectors, "*slices*", of width R (the transmission range).





The EBP Protocol (II)

- **Data Propagation:**

Each node in sector i propagates its messages according to the following rule:

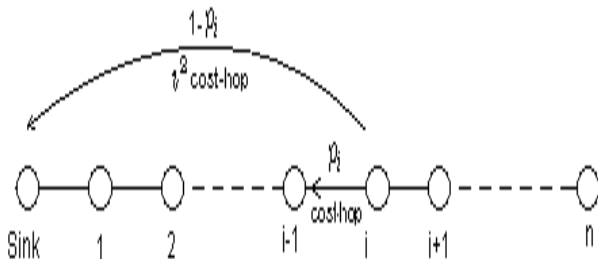
- Propagate the message to *sector $i - 1$* with *probability p_i* .
- Propagate the message *directly to the sink* with *probability $1 - p_i$* .

The *choice of p_i* is made such that the *average per sensor energy dissipation is the same* for all sensors in the network.



An Instance of the Execution

There is a message on node i .





Balanced Average per Sensor Energy Dissipation

- S_i : the *area size* of sector i .
- $\mathcal{E}_i$: the *total energy* dissipated in sector i .
- Sensor nodes are spread *uniformly at random* in the network area. We want:

$$\frac{E[\mathcal{E}_i]}{S_i} = \frac{E[\mathcal{E}_j]}{S_j} \quad \forall i, j \in \{1, \dots, n\}$$



The Model & Assumptions

The events occur in random uniform positions in the network.

Let ϵ_{ij} a *random variable* that measures the *dissipated energy* by sector i , so as to handle message j .

$$\epsilon_{ij} = \begin{cases} cR^2 & \text{with probability } p_i \\ c(iR)^2 & \text{with probability } 1 - p_i \end{cases}$$

Clearly,

$$E[\epsilon_{ij}] = p_i cR^2 + (1 - p_i) c(iR)^2 \Rightarrow$$

$$E[\epsilon_{ij}] = [i^2 - p_i(i^2 - 1)] cR^2$$



Computation of p_i (message handling)

- Let h_i the number of messages that were *handled* by sector i .
- Let f_i the number of messages that *were forwarded* to sector i .
- Let g_i the number of messages that *were generated* in sector i .

Clearly:

$$h_i = f_i + g_i$$

By linearity of expectation:

$$E[h_i] = E[f_i] + E[g_i]$$



Computation of p_i

Lemma

The following relationship holds:

$$E[f_i] = p_{i+1}E[h_{i+1}] = p_{i+1} \cdot (E[f_{i+1}] + E[g_{i+1}])$$

Clearly:

$$p_{i+1} = \frac{E[f_i]}{E[f_{i+1}] + E[g_{i+1}]}$$

The Energy Balance Property:

$$E \left[\frac{\sum_{k=1}^{h_i} \epsilon_{ik}}{S_i} \right] = E \left[\frac{\sum_{k=1}^{h_j} \epsilon_{jk}}{S_j} \right] \quad \forall i, j \in \{1, \dots, n\}$$



A recurrence relation for p_i

$$a(i+1)E[f_{i+1}] - (d(i) + a(i))E[f_i] + d(i-1)E[f_{i-1}] = a(i)E[g_i] - a(i+1)E[g_{i+1}]$$

where

$$a(i) = \frac{i^2}{2i-1} \quad d(i) = \frac{(i+1)^2 - 1}{2i+1}$$

Initial conditions:

$$E[f_n] = 0 \quad E[f_0] = n$$



Computation of p_i

Structure of the rest of the solution.

- First we transform the recurrence into a *simpler recurrence* with only *two successive terms* (whose coefficient is 1).
- Having solved the latter recurrency, we solve the initial one.
- Having computed the values of $E[f_i]$ for $i = 1, \dots, n$, we compute the exact values of probabilities p_i .



The exact solution

- The solution:

$$E[f_i] = - \sum_{k=1}^{n-i} \frac{\prod_{j=k}^{n-i+1} a(n-j)}{\prod_{j=k}^{n-i} d(n-j)} \cdot \left(\sum_{j=1}^{n-k} (a(j)E[g_j] - a(j+1)E[g_{j+1}]) + a(1) \cdot E[f_1] \right)$$

where

$$\prod_i^{i-1} a(i) = 1$$

- Easily computed in a repetitive manner
- Thus we can calculate p_i 's:

$$p_{i+1} = \frac{E[f_i]}{E[f_{i+1}] + E[g_{i+1}]}$$



A closed approximate form for p_i

If $E[f_i] \simeq E[f_{i-1}]$ then

$$p_i = 1 - \frac{3x}{(i+1)(i-1)} \quad \text{for } 3 \leq i \leq n$$

$p_2 = x$ and can be set to $\frac{1}{2}$

$E[f_i] \simeq E[f_{i-1}]$ is realistic because:

- $S_i \simeq S_{i-1}$
- $p_i \simeq p_{i-1}$.

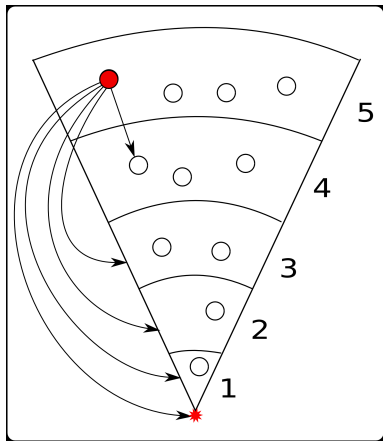


Comments on p_i

- when i is large, p_i is large, i.e. when far away from the sink it is better to move hop-by-hop, to avoid spending too much energy.
- when i becomes small, p_i is small, i.e. when we approach the sink it is better to transmit directly in order to bypass the critical region (and since energy consumption is small).



Lifespan Maximization



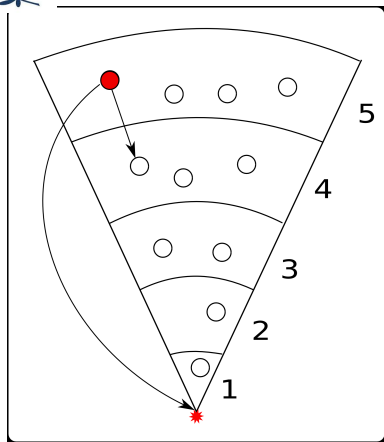
The algorithm balances energy consumption:

- Slicing of the network
- Generalized data propagation algorithm, allows to jump over bottle-neck nodes

Note: do hops to “intermediate” slices help?



Lifespan Maximization



Main Theoretical Result

Lifespan is maximized by a *mixed* data propagation algorithm

Application

This is used to propose a distributed **optimal** data propagation algorithm

- mixed propagation: only single hops and direct to sink transmissions
- mixed strategies beat every other possible strategy (wrt. lifespan)



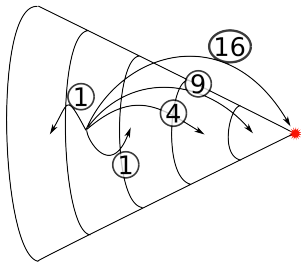
Model

① *Energy cost:*

Sending a message from slice i to j costs $(i - j)^2 E/\text{msg}$

② f_{ij} is the *message rate* from slice i slice to slice j msg/t

	<i>sink</i>	1	2	3	4
<i>sink</i>	0	1	4	9	16
1	1	0	1	4	9
2	4	1	0	1	4
3	9	4	1	0	1
4	16	9	4	1	0





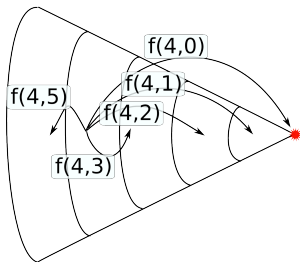
Model

① *Energy cost:*

Sending a message from slice i to slice j costs $(i - j)^2 E / \text{msg}$

② $f_{i,j}$ is the *message rate* from slice i slice to slice j msg/t

	<i>sink</i>	1	2	3	4
<i>sink</i>	$f_{0,0}$	$f_{0,1}$	$f_{0,2}$	$f_{0,3}$	$f_{0,4}$
1	$f_{1,0}$	$f_{1,1}$	$f_{1,2}$	$f_{1,3}$	$f_{1,4}$
2	$f_{2,0}$	$f_{2,1}$	$f_{2,2}$	$f_{2,3}$	$f_{2,4}$
3	$f_{3,0}$	$f_{3,1}$	$f_{3,2}$	$f_{3,3}$	$f_{3,4}$
4	$f_{4,0}$	$f_{4,1}$	$f_{4,2}$	$f_{4,3}$	$f_{4,4}$





Generalized-flow maximization

Problem

Given a WSN: Maximize the *generalized* network flow

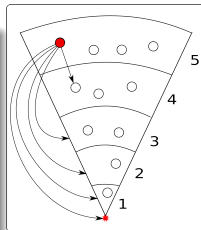
- Given the detection rates $f_{0,i} := g_i$
- Given the available energy: b_i

LP to maximize the generalized-flow:

Maximize T in:

- ① $f_{0,i} = Tg_i$ *Detection rates*
- ② $\sum_{j=0}^N f_{i,j}(i-j)^2 \leq b_i$ *Energy constraint*
- ③ $\sum_{j=0}^N f_{i,j} = \sum_{j=0}^N f_{j,i}$ *Flow equations*

- g_i : event generation rates (i.e. data injected at i)
- maximize time $T \Rightarrow$ lifespan maximization





Mixed-flow maximization

Problem

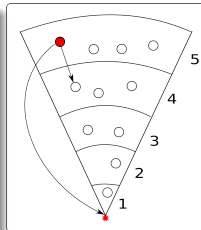
Given a WSN: Maximize the *mixed* network flow

- Given the detection rates $f_{0,i} := g_i$
- Given the available energy: b_i

LP to maximize the mixed-flow:

Maximize T in:

- 1 $f_{0,i} = Tg_i$ Detection rates
- 2 $\sum_{j=0}^N f_{i,j}(i-j)^2 \leq b_i$ Energy constraint
- 3 $\sum_{j=0}^N f_{i,j} = \sum_{j=0}^N f_{j,i}$ Flow equations
- 4 $f_{i,j} = 0$ if $j \notin \{0, i-1\}$ mixed flow constraint





Mixed-flow maximization

Problem

Given a WSN: Maximize the *mixed* network flow

- Given the detection rates $f_{0,i} := g_i$
- Given the available energy: b_i

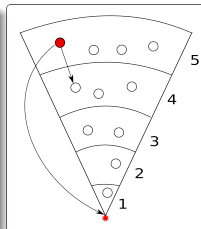
LP to maximize the mixed-flow:

Maximize T in:

- 1 $f_{0,i} = Tg_i$ *Detection rates*
- 2 $\sum_{j=0}^N f_{i,j}(i-j)^2 \leq b_i$ *Energy constraint*
- 3 $\sum_{j=0}^N f_{i,j} = \sum_{j=0}^N f_{j,i}$ *Flow equations*
- 4 $f_{i,j} = 0$ if $j \notin \{0, i-1\}$ *mixed flow constraint*

(4) guarantees no hops to “intermediate” slices

(3) guarantees flow preservation





Mixed-flows are optimal

Result

If there exists an NRG-balanced mixed flow

- 1 it maximizes the *mixed-flow*
- 2 it maximizes the *generalized-flow*

Application

We can maximize the generalized-flow problem using:

- a mixed-flow
- a local property (energy-balance)



A distributed algorithm

Definition

Inputs

- 1 Each node has a potential $potential(n) \simeq EnergySpent(n)$
- 2 Each node knows its list of neighbours $\mathcal{V}_n$
- 3 Each node knows the potential of its neighbours

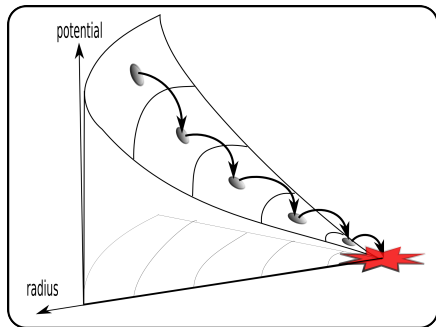
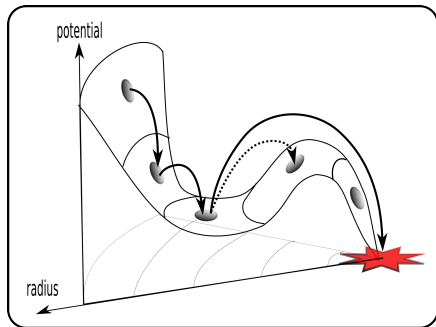
Algorithm Propagate Data

- Find m : the lowest potential neighbour
- If $potential(m) < potential(self)$ then send data to m
- Else send data directly to the sink



A distributed algorithm

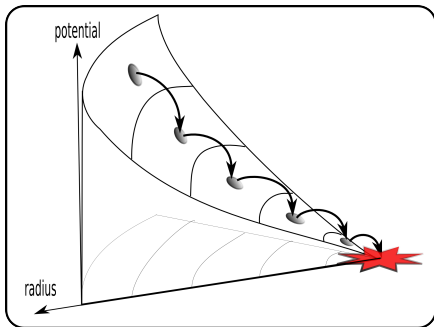
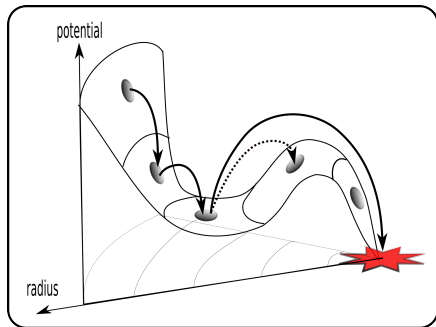
Illustration





A distributed algorithm

Illustration



Remark

- The algorithm produces a mixed flow
- The algorithm balances energy



Simulations

- Scatter nodes randomly over a region
- Events are generated randomly
- Data is propagated according to the algorithm



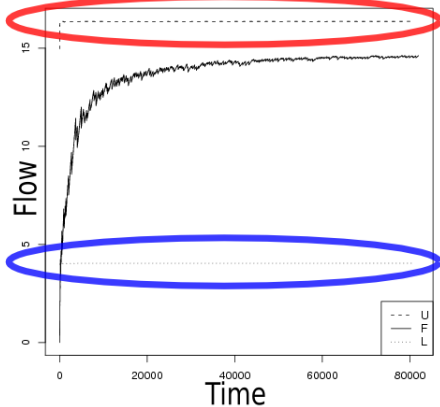
First Simulation

- 1000 sensors randomly dispersed over a 10m disc
- Sink at the center of the disc
- Potential function $potential(n) = Energy(n)$



First Simulation

Performances



Network flow

$$\frac{\text{Messages received by sink}}{\max_{\{nodes\}}(EnergySpent)}$$

F : the flow of the algorithm

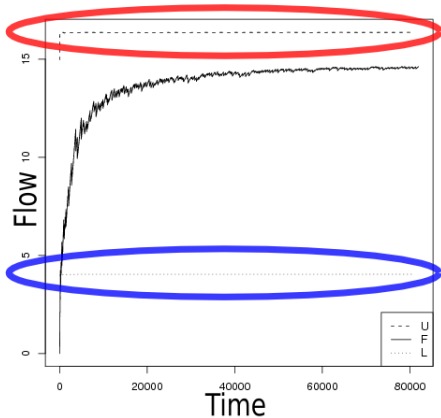
U : maximum possible flow (offline, computed by an LP)

L : maximum possible flow without direct transmissions



First Simulation

Performances

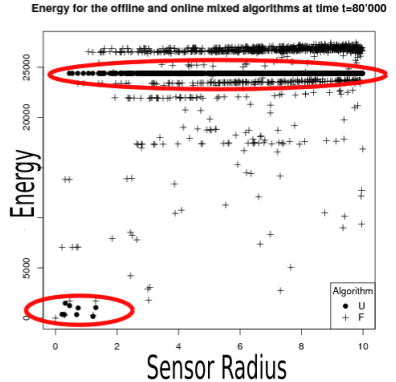
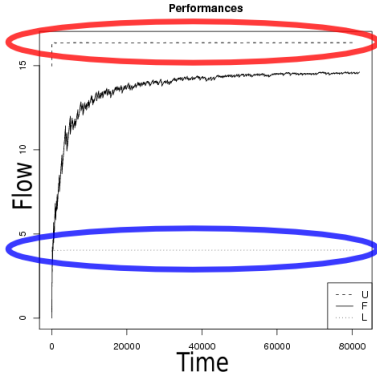


Remarks

- Close to optimal
- Distributed and on-line!



First Simulation

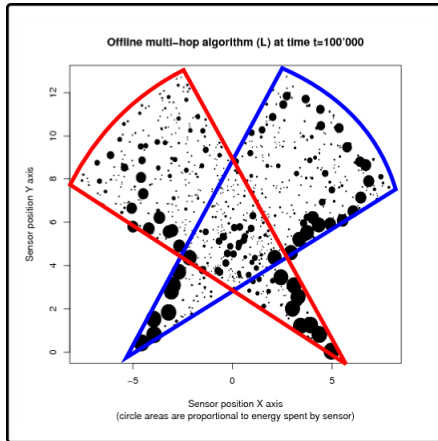


- off-line ideal flow U balances the energy load
- the online algorithm performs very well



Second Simulation

- 600 sensors randomly dispersed over 2 intersecting 30° sector graphs of 10m diameter with one sink at the narrow end of each sector
- events can be reported to either sink





Summary

- 1 Proved that optimal solution belongs to a subset of realistic data propagation algorithms: a *mixed strategy* which is *energy balanced*
- 2 Propose a distributed algorithm based on theoretical results
- 3 Show that the algorithm is efficient (Markov chain context and simulations)



References

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